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STUDY AND EXPERIMENTAL EVALUATION OF AN SDR-BASED DVB-T TRANSMISSION/RECEPTION CHAIN

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Abstract

This report presents the study and experimental evaluation of a Digital Video Broadcasting – Terrestrial (DVB-T) transmission and reception chain implemented using Software Defined Radio (SDR) technology. The system was developed using GNU Radio Companion and the NI USRP 2920 software defined radio platform. Two modulation schemes were investigated: 16-Quadrature Amplitude Modulation (16-QAM) and 64-QAM, both operating in 8K OFDM mode with a code rate of 2/3. The 16-QAM chain was successfully implemented and validated both in simulation and with real SDR hardware, producing clean constellation diagrams and a successfully recovered MPEG transport stream at the receiver. The 64-QAM chain was validated in simulation but encountered significant underflow and overflow issues during real hardware transmission due to the increased computational load and hardware limitations. This work demonstrates the feasibility of implementing a complete DVB-T broadcast chain using open-source SDR tools, while highlighting the real-world practical challenges encountered during experimental deployment.

Project Objective

The objective of this project is to design, implement, and experimentally evaluate a complete DVB-T transmission and reception chain using Software Defined Radio (SDR) technology. This work aims to:

- Understand the principles of DVB-T and OFDM transmission.
- Implement the full physical layer using GNU Radio and the `gr-dvbt` library.
- Validate the system through simulation and real SDR hardware experiments.
- Analyze the impact of modulation schemes (16-QAM and 64-QAM).
- Evaluate system limitations such as hardware constraints and computational load.
- Investigate real-time transmission capabilities using UDP streaming, highlighting the challenges of implementing DVB-T systems in practical environments.

1. Background on DVB-T and OFDM Transmission

1.1 The DVB-T Standard

Digital Video Broadcasting – Terrestrial (DVB-T) is an international standard for terrestrial digital television, developed by the DVB Project and published by the European Telecommunications Standards Institute in 1997. It has become one of the most widely adopted systems worldwide, used in over 120 countries.

DVB-T replaced analog TV broadcasting by improving spectral efficiency, transmission robustness, and picture quality. It supports services such as SDTV, HDTV, radio, and interactive applications, and enables reception on fixed, portable, and mobile devices.

1.2 DVB-T Key Parameters

DVB-T is a highly flexible standard that allows network operators to configure transmission parameters according to their specific requirements, finding the right balance between robustness and capacity. The key configurable parameters are summarized in Table 1.

Table 1: DVB-T Configurable Parameters

Parameter	Options
Modulation	QPSK, 16-QAM, 64-QAM
FFT Size	2K, 8K
Guard Interval	1/4, 1/8, 1/16, 1/32
Code Rate	1/2, 2/3, 3/4, 5/6, 7/8
Channel Bandwidth	6, 7, or 8 MHz

In this project, the following parameter set was used:

Table 2: Parameters Used in This Project

Parameter	Value
Modulation	16-QAM / 64-QAM
FFT Size	8K
Guard Interval	1/32
Code Rate	2/3
Hierarchy	Non-Hierarchical

1.3 OFDM Transmission Principle

Orthogonal Frequency Division Multiplexing (OFDM) is the modulation technique used in DVB-T. OFDM divides the available channel bandwidth into a large number of closely spaced orthogonal subcarriers, each carrying a small portion of the total data. This approach offers several key advantages:

- **Robustness against multipath interference:** The long OFDM symbol duration makes the system resilient to echoes and reflections.
- **Efficient spectrum use:** Orthogonal subcarriers can overlap without interference.
- **Single Frequency Network (SFN) support:** Multiple transmitters can operate on the same frequency.

In 8K mode, DVB-T uses 8192 subcarriers, of which 6817 carry useful data (occupied tones). The remaining subcarriers are used as guard bands, pilot signals, and reference signals for synchronization and channel estimation.

1.4 The Guard Interval

The guard interval is a cyclic prefix added at the beginning of each OFDM symbol. It protects against inter-symbol interference caused by multipath propagation. In this project, a guard interval of 1/32 was used, meaning the cyclic prefix length is 1/32 of the FFT size:

$$\text{CP Length} = \frac{8192}{32} = 256 \text{ samples} \quad (1)$$

1.5 16-QAM vs. 64-QAM Modulation

Quadrature Amplitude Modulation (QAM) encodes data by varying both the amplitude and phase of the carrier signal. The key difference between 16-QAM and 64-QAM is the number of constellation points, summarized in Table 3.

Table 3: Comparison of 16-QAM and 64-QAM

Modulation	Constellation Points	Bits/Symbol	Robustness	Capacity
16-QAM	16 (4×4 grid)	4 bits	Higher	Lower
64-QAM	64 (8×8 grid)	6 bits	Lower	Higher

64-QAM offers 50% more capacity than 16-QAM but requires a significantly higher Signal-to-Noise Ratio (SNR) for reliable reception, making it more sensitive to channel impairments and hardware limitations.

1.6 Channel Coding in DVB-T

DVB-T employs a concatenated channel coding scheme to protect the transmitted data against errors:

- **Outer coding – Reed-Solomon (RS):** The Reed-Solomon encoder adds redundancy to protect against burst errors. The parameters used are RS(255, 239), meaning 255 bytes are transmitted for every 239 bytes of data, with the ability to correct up to 8 byte errors per codeword.
- **Outer interleaving – Convolutional Interleaver:** The convolutional interleaver rearranges the data bytes to spread burst errors across multiple codewords, improving the effectiveness of the Reed-Solomon decoder.
- **Inner coding – Convolutional Code:** A punctured convolutional code with code rate 2/3 adds further error protection at the bit level.
- **Inner interleaving – Bit and Symbol Interleaving:** Two stages of interleaving (bit inner interleaver and symbol inner interleaver) further distribute errors to improve system robustness.

1.7 Pilots and Reference Signals

DVB-T inserts pilot subcarriers throughout the OFDM frame. These pilots are used by the receiver for:

- **Synchronization:** Time and frequency synchronization.

- **Channel estimation:** Measuring the channel response.
- **Equalization:** Correcting channel distortions.

The *Demod Reference Signals* block at the receiver uses these pilots to perform channel equalization, which is why the equalized constellation appears as a clean 16-QAM grid even after passing through an imperfect channel.

2. Description of the Implemented SDR Chain

2.1 End-to-End System Operation

The DVB-T transmission and reception chain can be understood as a sequence of signal transformations from the original data source to the final reconstructed output.

At the transmitter side, the MPEG transport stream is first processed through energy dispersal to randomize the data and avoid spectral peaks. The signal is then protected against errors using Reed-Solomon encoding and convolutional coding. Interleaving stages are applied to distribute burst errors over time and frequency.

The encoded bitstream is then mapped onto complex symbols using QAM modulation. These symbols are inserted into an OFDM frame, where each subcarrier carries a portion of the data. An inverse FFT converts the signal into the time domain, and a cyclic prefix is added to mitigate inter-symbol interference.

The signal is transmitted through the wireless channel using the USRP hardware.

At the receiver side, the process is reversed. The received signal undergoes synchronization, FFT processing, and channel estimation. The constellation is equalized before demapping the symbols back into bits. Deinterleaving and decoding stages recover the original data stream.

Finally, the MPEG transport stream is reconstructed and written to an output file, confirming the correct operation of the system.

This end-to-end process highlights the importance of each processing block and demonstrates how digital communication systems ensure reliable data transmission over imperfect channels.

2.2 Hardware Platform – NI USRP 2920

The NI USRP 2920 is a software defined radio platform designed for wireless communications research and education. Its key specifications are given in Table 4.

Table 4: NI USRP 2920 Specifications

Specification	Value
Frequency Range	50 MHz – 2.2 GHz
Master Clock Rate	100 MHz
Interface	Gigabit Ethernet
TX Port	RX / TX
RX Port	RX2
Max TX Output	+20 dBm
Max RX Input	−15 dBm

2.3 Software Platform – GNU Radio and gr-dvbt

GNU Radio Companion (GRC) is an open-source software framework for signal processing and SDR development. The `gr-dvbt` library provides a complete implementation of the DVB-T physical layer for GNU Radio, including all encoding, modulation, and decoding blocks.

2.4 System Configuration Parameters

The following parameters were used in the final working configuration:

Table 5: System Configuration Parameters

Parameter	Value	Justification
Sample Rate	4 M	Compatible (PC capability)
Center Frequency	2.48 GHz	Characteristics of the antennas used
Master Clock Rate	100 MHz	Fixed USRP 2920 clock
Bandwidth	4 M	Equal to sample rate
TX Gain	1 (normalized)	Optimal for hardware
RX Gain	500 m	Optimal for hardware
FFT Size	8192 (8K)	DVB-T standard
Guard Interval	1/32	DVB-T standard
Code Rate	2/3	DVB-T standard
Modulation	16-QAM	Best stability

2.5 MPEG Transport Stream (MPEG-TS)

The MPEG Transport Stream (MPEG-TS) is a standard format used for transmitting audio, video, and data in digital broadcasting systems such as DVB-T. It is specifically designed for reliable transmission

over noisy or unreliable communication channels.

Each MPEG-TS packet has a fixed length of 188 bytes, which allows efficient synchronization and error detection. The structure of a packet includes:

- **Sync Byte (0x47):** Identifies the start of a packet.
- **Header:** Contains information such as packet ID (PID), continuity counter, and error indicators.
- **Payload:** Contains the actual data (video, audio, or metadata).

The fixed packet size simplifies buffering and processing in both transmission and reception systems. It also allows easy detection of lost or corrupted packets.

In this project, the file `test_short.ts` is used as the input source and continuously streamed through the DVB-T transmission chain. At the receiver, the decoded stream is written into `test_out.ts`.

The comparison between the input and output files confirms that the transmission chain preserves data integrity. This validates the correct operation of encoding, modulation, transmission, and decoding stages.

The use of MPEG-TS ensures compatibility with real broadcasting systems and demonstrates the practical applicability of the implemented DVB-T chain.

2.6 Transmission Chain (TX)

The transmission chain processes the input MPEG transport stream through the following stages:

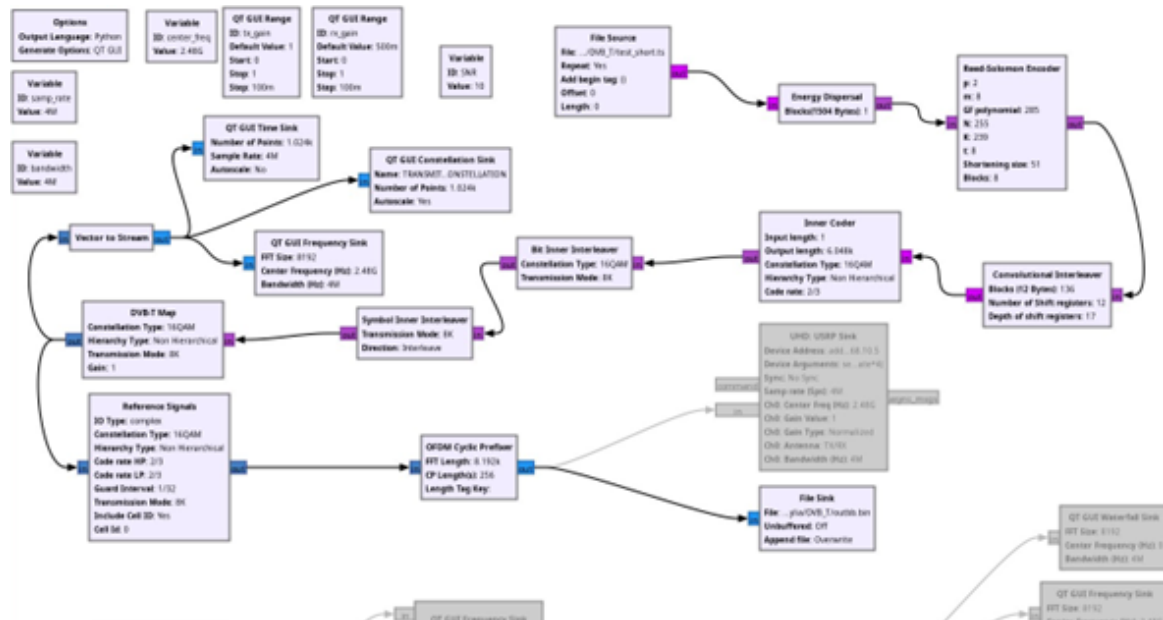


Figure 1: GNU Radio flowgraph of the DVB-T transmission chain (TX).

1. **File Source** – Reads the input MPEG transport stream file (`test_short.ts`) and feeds it into the processing chain. The file is read continuously with *Repeat* enabled to ensure continuous transmission.
2. **Energy Dispersal** – Randomizes the input data stream to ensure a uniform power spectral density. This prevents the transmission of long sequences of identical bits which would cause spectral peaks. Block size: 1504 bytes.

3. **Reed-Solomon Encoder** – Adds outer error correction using RS(255, 239, $t=8$) code with GF polynomial 285. For every 239 bytes of data, 16 redundancy bytes are added, allowing correction of up to 8 byte errors.
4. **Convolutional Interleaver** – Interleaves the RS-encoded data using 12 branches of depth 17 to spread burst errors across multiple codewords.
5. **Inner Coder** – Applies a punctured convolutional code with rate $2/3$, adding inner error protection at the bit level. The output length is 6.048k bits.
6. **Bit Inner Interleaver** – Interleaves the encoded bits to further distribute errors. Configured for 16-QAM, Transmission Mode 8K.
7. **Symbol Inner Interleaver** – Performs symbol-level interleaving in the Interleave direction for Transmission Mode 8K.
8. **DVB-T Map** – Maps the interleaved bits to 16-QAM constellation points. Each group of 4 bits is mapped to one of 16 complex symbols arranged in a 4×4 grid.
9. **Reference Signals** – Inserts pilot subcarriers and reference signals into the OFDM frame. These signals are used by the receiver for synchronization and channel equalization.
10. **OFDM Cyclic Prefixer** – Performs the IFFT to convert the frequency-domain OFDM symbols to the time domain, and adds the cyclic prefix of 256 samples. FFT Length: 8192.
11. **UHD: USRP Sink** – Transmits the baseband signal through the USRP 2920 hardware via the RX1/TX antenna port at 2.48 GHz center frequency.

Simulation Results – Transmission Chain

To evaluate the performance of the DVB-T transmission chain, a simulation was conducted. The following figures illustrate the time-domain signals and constellation diagrams obtained at different stages of the system.

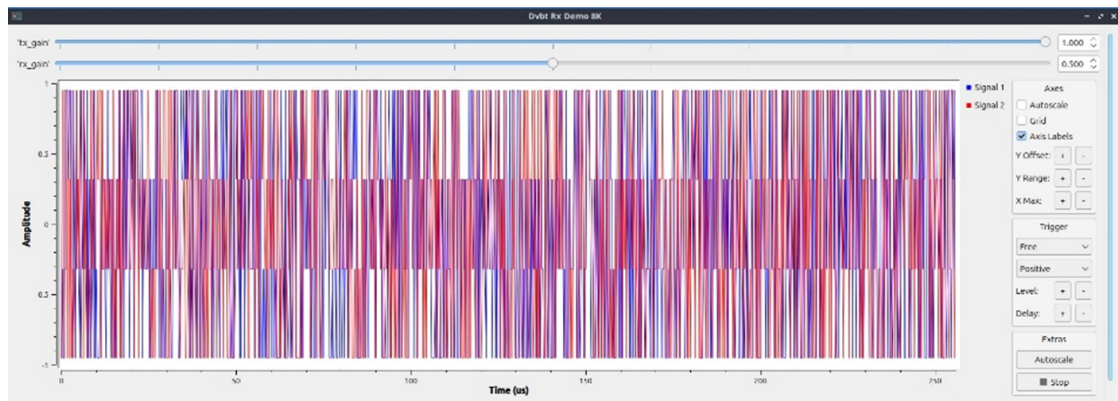


Figure 2: Time-domain representation of the transmitted DVB-T signal.

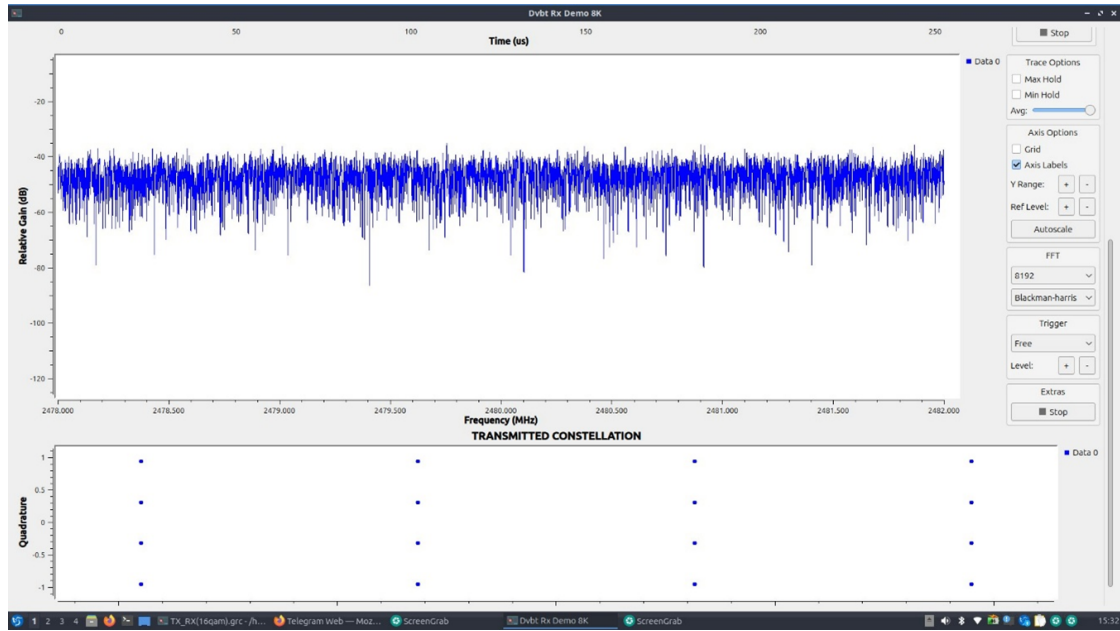


Figure 3: Time-domain representation of the received DVB-T signal (top) and corresponding constellation diagram (bottom).

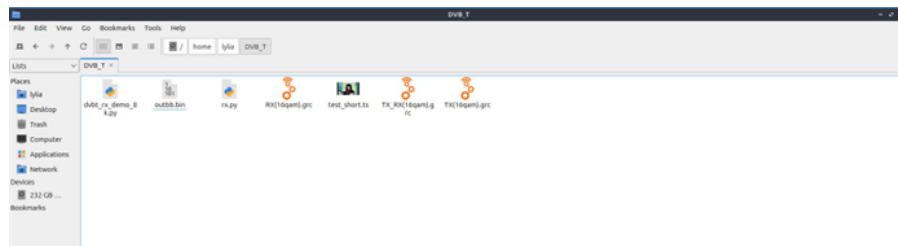


Figure 4: Created output file in simulation of transmission.

Key observations from the transmission chain:

- Each block plays a critical role in ensuring reliable communication.
- Energy dispersal prevents spectral peaks and ensures uniform signal distribution.
- Reed-Solomon coding protects against burst errors, while convolutional coding adds robustness at the bit level.
- Interleaving spreads errors across the data stream, making them easier to correct at the receiver.
- OFDM modulation enables efficient spectrum usage and robustness against multipath propagation.

2.7 Reception Chain (RX)

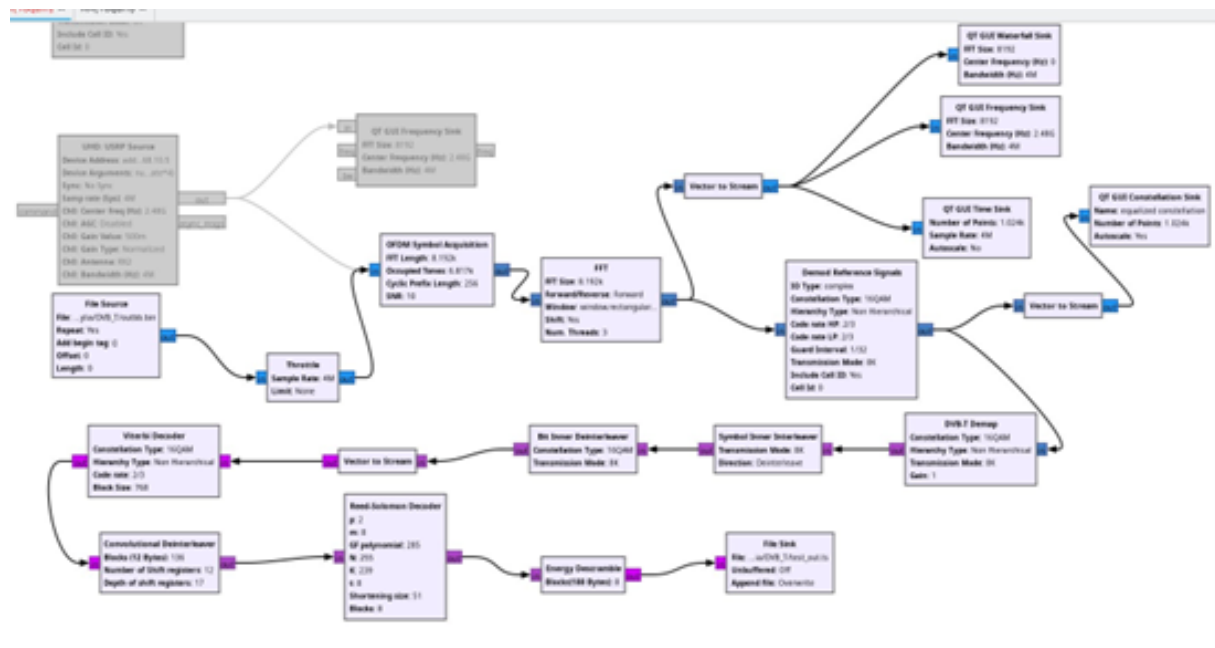


Figure 5: GNU Radio flowgraph of the DVB-T reception chain (RX).

1. **UHD: USRP Source** – Receives the RF signal through the USRP 2920 via the RX2 antenna port. AGC is disabled and gain is manually set.
2. **OFDM Symbol Acquisition** – Performs time synchronization and detects the start of each OFDM symbol using the cyclic prefix. FFT Length: 8192, Occupied Tones: 6817, Cyclic Prefix Length: 256, SNR: 10.
3. **FFT** – Converts the received time-domain signal to the frequency domain using an 8192-point FFT (Forward direction).
4. **Demod Reference Signals** – Uses the received pilot subcarriers to perform channel estimation and equalization. This block corrects the phase and amplitude distortions introduced by the channel, producing a clean equalized constellation.
5. **DVB-T Demap** – Demaps the equalized constellation points back to bit sequences. Each 16-QAM symbol is converted to 4 bits.
6. **Symbol Inner Deinterleaver and Bit Inner Deinterleaver** – Reverses the interleaving applied at the transmitter.
7. **Viterbi Decoder** – Decodes the convolutional code using the Viterbi algorithm with Block Size 768.
8. **Convolutional Deinterleaver** – Reverses the convolutional interleaving.
9. **Reed-Solomon Decoder** – Corrects remaining errors using the RS(255, 239, $t=8$) decoder.
10. **Energy Descramble** – Reverses the energy dispersal to recover the original data stream. Block size: 188 bytes.
11. **File Sink** – Writes the recovered transport stream to `test_out.ts`.

Simulation Results – Reception Chain

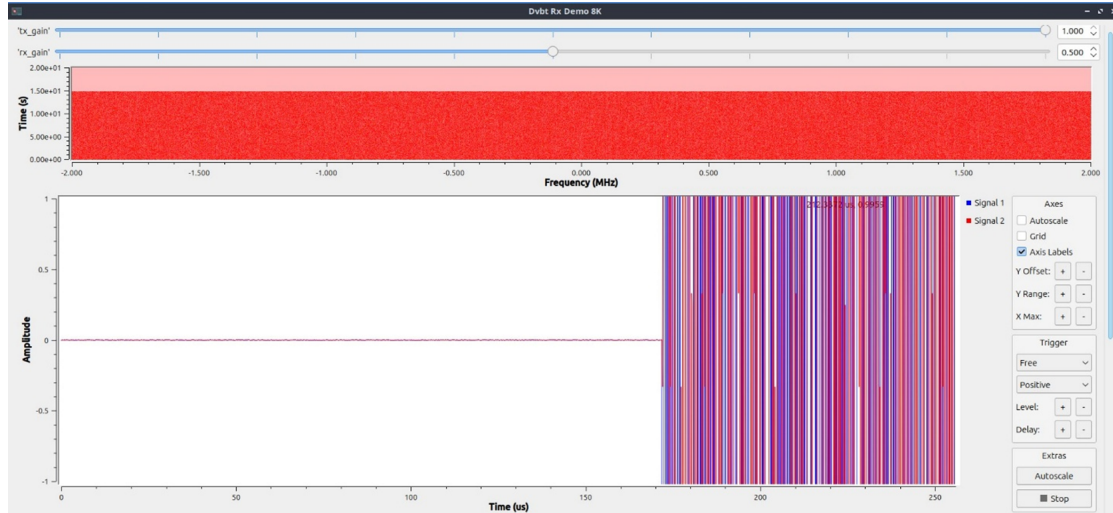


Figure 6: Waterfall spectrum (top) and time-domain received signal (bottom).

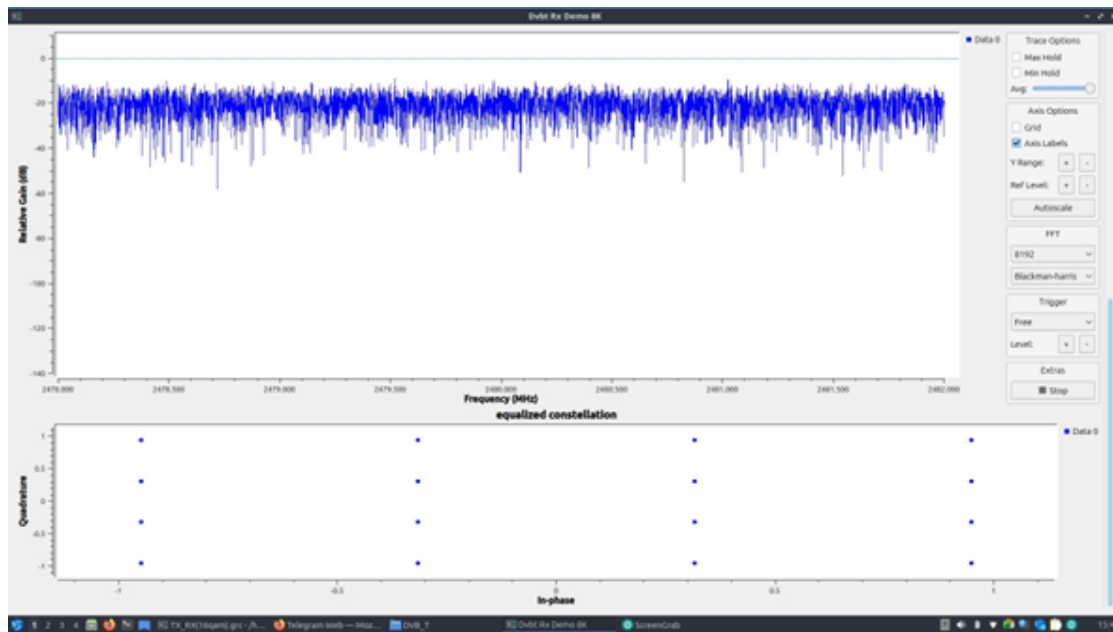


Figure 7: Frequency spectrum and equalized constellation of the received signal.



Figure 8: Output file (received video) after successful decoding.

Real Experiment Using SDR Hardware

To validate the DVB-T transmission chain under real conditions, both the transmitter (TX) and receiver (RX) flowgraphs were combined using SDR hardware. The following results present the received signal in time, frequency, and constellation domains.

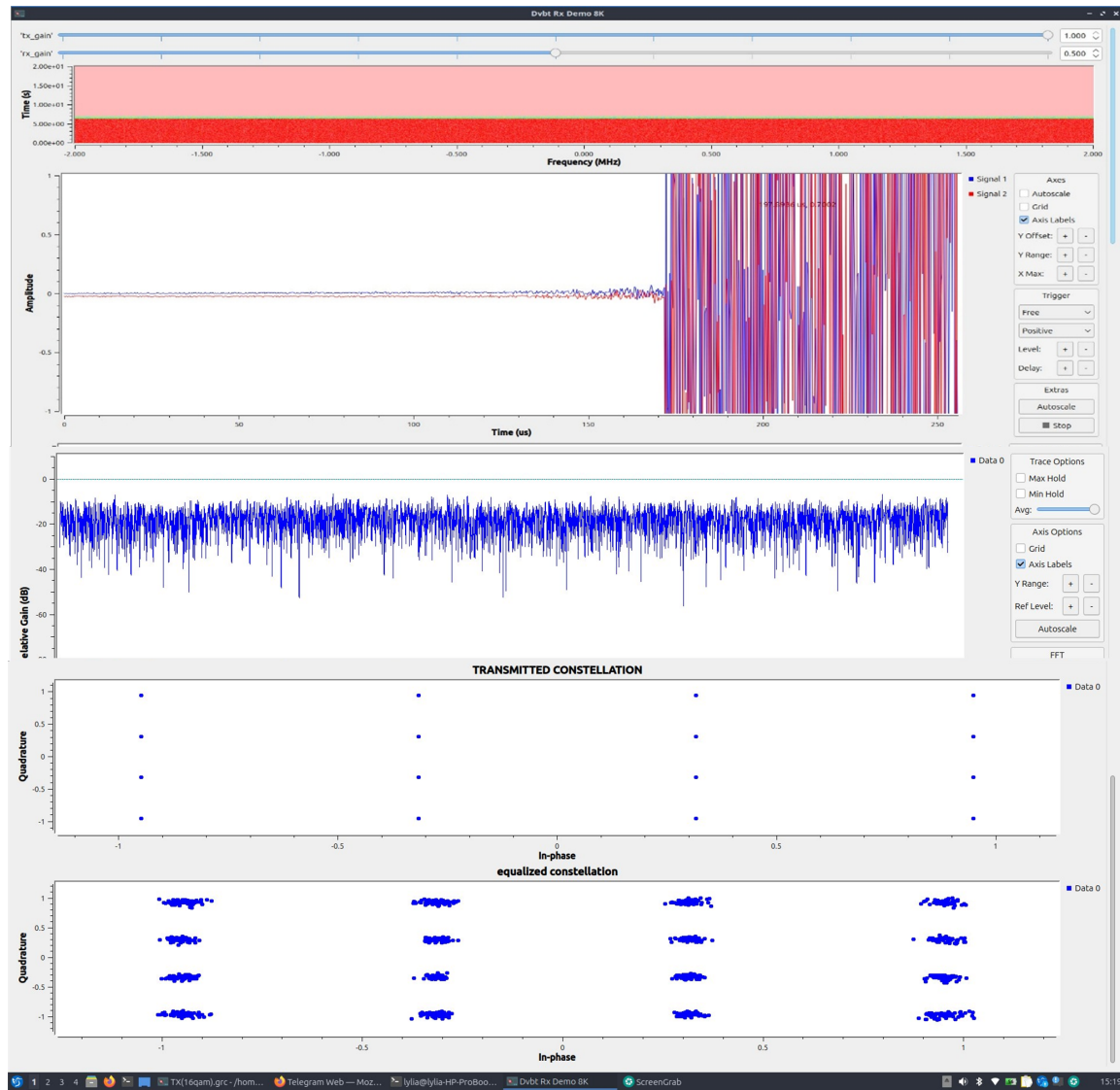


Figure 9: Real-time SDR experiment showing waterfall spectrum, time-domain signal, frequency spectrum, and transmitted/equalized constellation diagrams.

Key observations from the reception chain:

- Synchronization is a critical step, as accurate timing and frequency alignment are required to correctly recover OFDM symbols.
- Channel estimation and equalization compensate for distortions introduced by the transmission channel.
- Demapping and decoding reconstruct the original bitstream, while error correction mechanisms ensure data integrity.
- The successful reconstruction of the output MPEG-TS file confirms the effectiveness of the receiver chain.

3. Complete DVB-T System Block Diagram

The complete DVB-T transmission and reception chain is illustrated in Figure 10. This diagram summarizes all processing stages from the input MPEG transport stream to the reconstructed output file, including the wireless channel traversed by the SDR hardware.

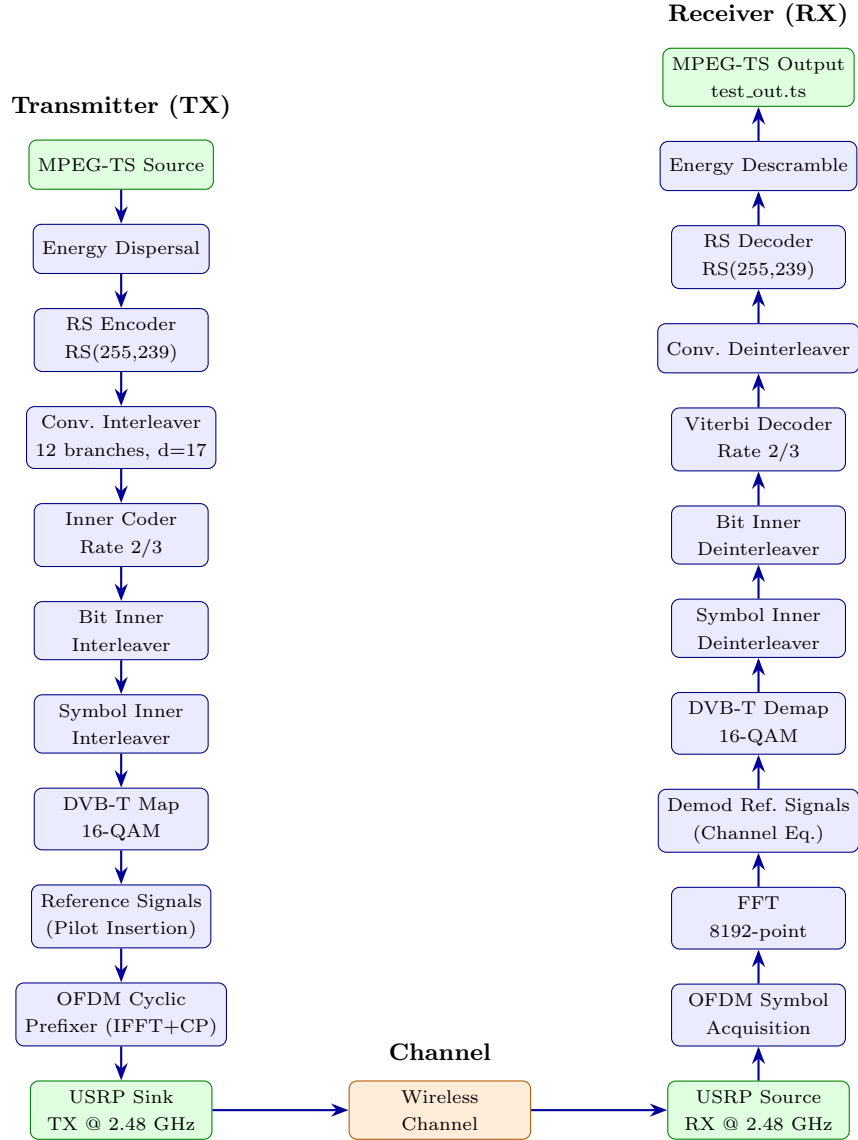


Figure 10: Complete block diagram of the SDR-based DVB-T transmission/reception chain.

The diagram highlights the symmetry between the transmitter and receiver chains. Each processing block at the transmitter has a corresponding inverse block at the receiver. The wireless channel introduces noise, multipath fading, and frequency offsets, all of which are compensated by the synchronization and equalization stages at the receiver.

4. Modifications Introduced

Several modifications were applied to adapt the DVB-T chain for stable operation:

- Adjustment of the sample rate to 4 MHz to match hardware capabilities.
- Configuration of the master clock rate to 100 MHz (USRP constraint).

- Manual gain control (TX and RX) to avoid signal saturation – disabling AGC for better stability.
- Integration of TX and RX flowgraphs for real-time testing.
- Addition of visualization tools (waterfall, constellation, spectrum).
- Optimization of parameters for stable 16-QAM transmission.

These modifications were essential to ensure proper system performance in both simulation and real hardware environments.

5. 64-QAM Performance Analysis

The previous sections validated the DVB-T system using 16-QAM modulation in both simulation and real hardware. In this section, a higher-order modulation scheme (64-QAM) is investigated to evaluate its performance and limitations under the same conditions.

As an additional investigation, the 16-QAM chain was modified to operate with 64-QAM modulation. All relevant blocks (DVB-T Map, Bit Inner Interleaver, Inner Coder, DVB-T Demap, and Viterbi Decoder) were reconfigured accordingly.

Simulation results showed correct operation, with a well-defined constellation composed of 64 distinct points arranged in an 8×8 grid.

However, real hardware implementation revealed significant limitations. The higher-order modulation is more sensitive to noise, synchronization errors, and channel imperfections, leading to dispersion of constellation points. Additionally, the increased data rate introduced processing constraints, resulting in timing issues and occasional buffer overflows on the USRP. Consequently, stable real-time transmission with 64-QAM was difficult to achieve.

5.1 DVB-T Transmission and Reception Chain (64-QAM)

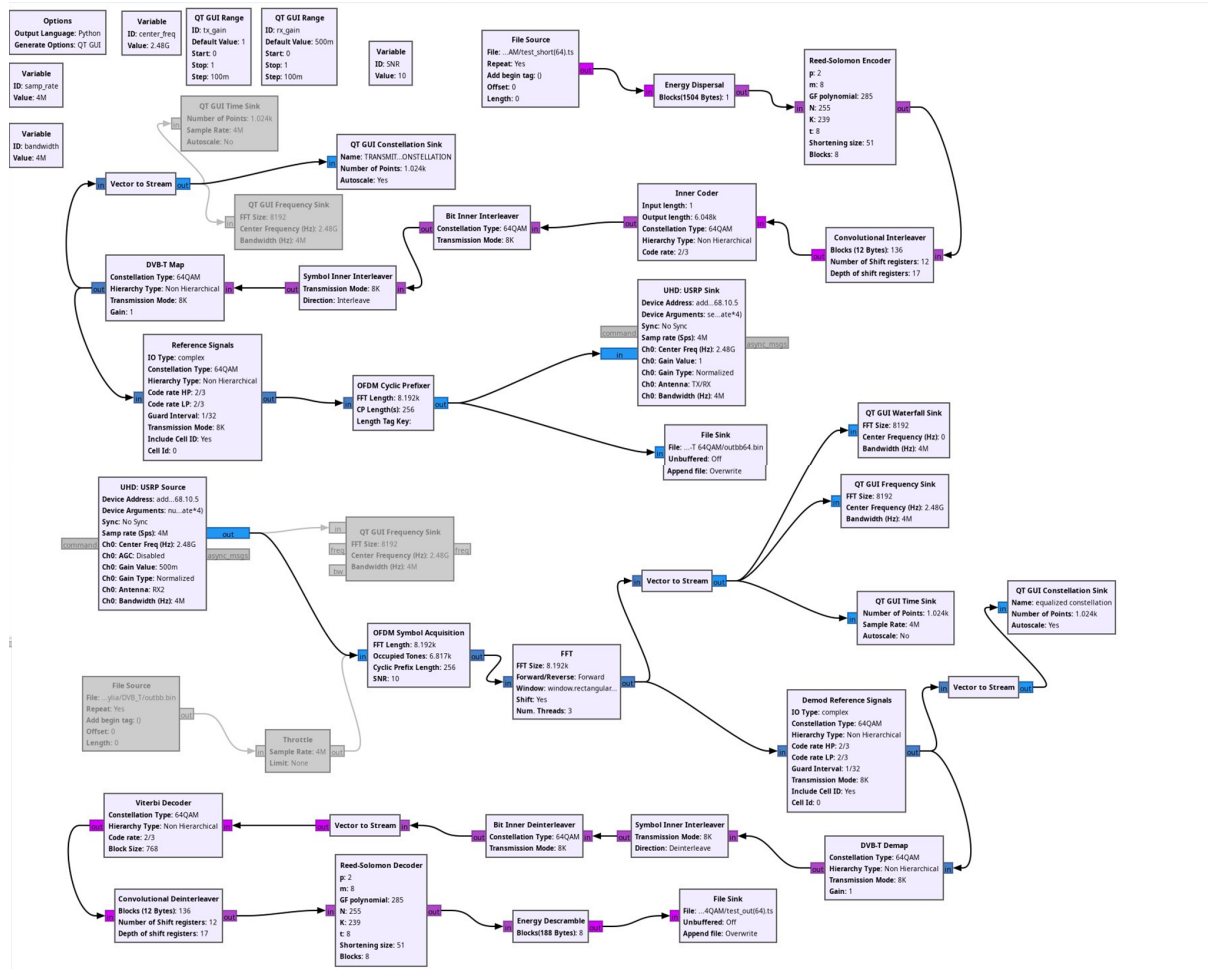


Figure 11: GNU Radio flowgraph of the DVB-T 64-QAM transmission and reception chain.

Simulation Results – 64-QAM

To validate the 64-QAM chain before hardware testing, a file-based simulation was conducted. The transmitter read the input transport stream, applied 64-QAM encoding through all DVB-T processing stages, and wrote the intermediate baseband signal to a file. The receiver then processed this file and recovered the transport stream.

The simulation produced a well-defined 64-QAM constellation composed of 64 distinct points arranged in an 8×8 grid at both the transmitter and receiver, confirming that the 64-QAM chain was correctly configured at the software level.

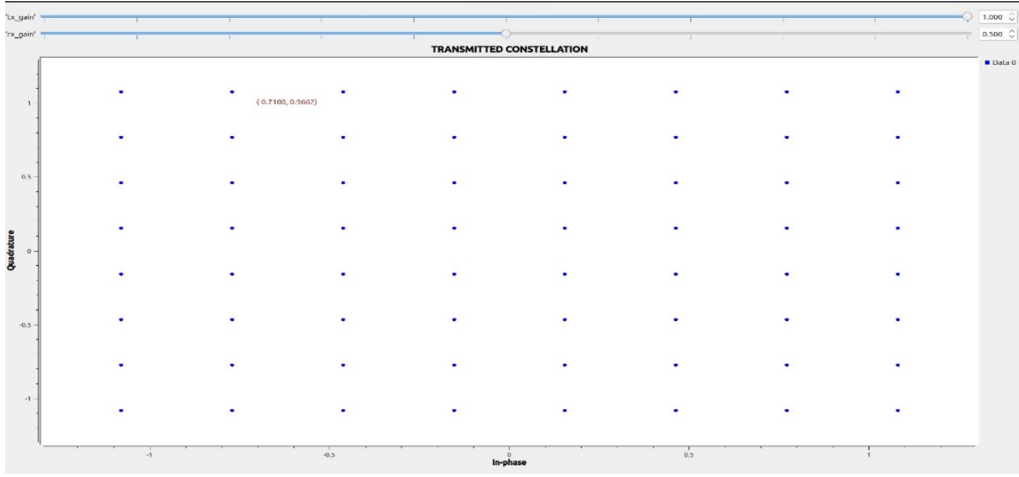


Figure 12: 64-QAM transmission constellation (simulation).



Figure 13: 64-QAM reception output (simulation).

5.2 Real Hardware Results (64-QAM)

When the 64-QAM chain was tested with the USRP 2920 hardware, several critical issues were encountered:

- **Underflow errors:** `usrp_sink: error: In the last 8281 ms, 940 underflows occurred.`
- The host PC could not generate samples fast enough to continuously feed the USRP, causing signal discontinuities. This is because 64-QAM with 8K OFDM requires significantly more processing power than 16-QAM – each symbol carries 6 bits instead of 4, increasing the computational load by 50%.
- **Overflow errors:** The receiver also experienced overflow conditions, where the USRP produced

samples faster than the PC could process them, resulting in lost data packets.

5.3 Root Cause Analysis

Table 6: Root Cause Analysis for 64-QAM Hardware Failures

Factor	Impact on 64-QAM
Computational load	50% higher than 16-QAM
Required SNR	Much higher than 16-QAM
Windows scheduling	Limited real-time performance
PC processing speed	Insufficient for 64-QAM + 8K OFDM

5.4 Comparison: 16-QAM vs. 64-QAM

Table 7: 16-QAM vs. 64-QAM Comparison

Aspect	16-QAM	64-QAM
Bits per symbol	4	6
Simulation	Perfect	Perfect
Real hardware	Stable	Underflows
Computational load	Moderate	High
Required SNR	~16 dB	~22 dB
Robustness	Higher	Lower

6. Theoretical BER Analysis

The Bit Error Rate (BER) is one of the most important metrics to evaluate the performance of a digital communication system. It represents the probability that a received bit is in error. For QAM modulation over an Additive White Gaussian Noise (AWGN) channel, the theoretical BER can be approximated analytically.

6.1 BER Expression for 16-QAM

For 16-QAM, the theoretical BER over an AWGN channel is given by:

$$P_{b, 16\text{-QAM}} = \frac{3}{8} \cdot \text{erfc} \left(\sqrt{\frac{E_b}{5 N_0}} \right) \quad (2)$$

where $\text{erfc}(\cdot)$ is the complementary error function, E_b is the energy per bit, and N_0 is the one-sided noise power spectral density.

6.2 BER Expression for 64-QAM

For 64-QAM, the theoretical BER over an AWGN channel is approximated by:

$$P_{b, 64\text{-QAM}} \approx \frac{7}{24} \cdot \text{erfc} \left(\sqrt{\frac{E_b}{21 N_0}} \right) \quad (3)$$

These expressions clearly show that 64-QAM requires a significantly higher E_b/N_0 ratio to achieve the same BER as 16-QAM. The table below summarizes the minimum E_b/N_0 required to reach a BER of 10^{-6} under ideal AWGN conditions:

Table 8: Required E_b/N_0 for BER = 10^{-6} (AWGN, code rate 2/3)

Modulation	Required E_b/N_0 (dB)	Minimum SNR (dB)
16-QAM	≈ 14.4	≈ 16.0
64-QAM	≈ 20.5	≈ 22.0

6.3 Effect of Channel Coding on BER

The concatenated coding scheme used in DVB-T (Reed-Solomon outer code + punctured convolutional inner code at rate 2/3) significantly improves the effective BER performance. The convolutional code provides a coding gain of approximately 5–6 dB, while the Reed-Solomon outer code eliminates residual burst errors after Viterbi decoding.

As a result, the minimum SNR required for quasi-error-free reception (defined as BER < 10^{-11} after Reed-Solomon decoding) is:

$$\text{SNR}_{\min, 16\text{-QAM, coded}} \approx 16.0 \text{ dB} \quad (4)$$

$$\text{SNR}_{\min, 64\text{-QAM, coded}} \approx 22.0 \text{ dB} \quad (5)$$

In our real hardware experiment with 16-QAM, the system operated above this SNR threshold, which explains the successful recovery of the MPEG transport stream. In contrast, the 64-QAM hardware experiment suffered from additional SNR degradation caused by underflow-induced signal discontinuities, effectively pushing the received SNR below the minimum required threshold.

6.4 Spectral Efficiency Comparison

The spectral efficiency η of a coded DVB-T signal is given by:

$$\eta = \log_2(M) \times R_c \times \frac{N_{\text{data}}}{N_{\text{total}}} \times \frac{T_u}{T_u + T_{GI}} \quad [\text{bit/s/Hz}] \quad (6)$$

where M is the modulation order, $R_c = 2/3$ is the code rate, $N_{\text{data}} = 6817$ is the number of data subcarriers, $N_{\text{total}} = 8192$ is the total FFT size, and $T_{GI}/T_u = 1/32$ is the guard interval ratio.

Applying this formula to our project parameters at 4 MHz bandwidth:

Table 9: Spectral Efficiency and Achievable Bit Rate for Our System Parameters

Modulation	Bits/Symbol	Spectral Eff. (bit/s/Hz)	Bit Rate at 4 MHz
16-QAM	4	≈ 2.22	≈ 8.9 Mbit/s
64-QAM	6	≈ 3.33	≈ 13.3 Mbit/s

This confirms that 64-QAM offers a 50% increase in throughput over 16-QAM, at the cost of stricter SNR requirements and higher computational load, as experimentally demonstrated in this project.

7. Useful Bit Rate Calculation

This analysis complements the spectral efficiency results by providing a concrete evaluation of the achievable data rate in our system.

A key advantage of the DVB-T standard is its flexible parameter set, which allows the exact useful bit rate to be computed analytically from system parameters. This section derives the theoretical throughput for both 16-QAM and 64-QAM under our experimental configuration.

7.1 OFDM Symbol Duration

The total OFDM symbol duration T_s is the sum of the useful symbol duration T_u and the guard interval T_{GI} :

$$T_u = \frac{N_{FFT}}{f_s} = \frac{8192}{4 \times 10^6} = 2.048 \text{ ms} \quad (7)$$

$$T_{GI} = \frac{T_u}{32} = \frac{2.048}{32} = 64 \text{ } \mu\text{s} \quad (8)$$

$$T_s = T_u + T_{GI} = 2.048 + 0.064 = 2.112 \text{ ms} \quad (9)$$

Therefore the OFDM symbol rate is:

$$R_{\text{symbol}} = \frac{1}{T_s} = \frac{1}{2.112 \times 10^{-3}} \approx 473.5 \text{ symbols/s} \quad (10)$$

7.2 Raw Bit Rate per Subcarrier

Each occupied subcarrier carries $\log_2(M)$ bits per OFDM symbol, where M is the constellation size. With $N_{\text{data}} = 6817$ data-carrying subcarriers:

$$R_{\text{raw, 16-QAM}} = N_{\text{data}} \times \log_2(16) \times R_{\text{symbol}} = 6817 \times 4 \times 473.5 \approx 12.93 \text{ Mbit/s} \quad (11)$$

$$R_{\text{raw, 64-QAM}} = N_{\text{data}} \times \log_2(64) \times R_{\text{symbol}} = 6817 \times 6 \times 473.5 \approx 19.39 \text{ Mbit/s} \quad (12)$$

7.3 Net Useful Bit Rate (After Channel Coding)

After applying the convolutional code at rate $R_c = 2/3$, the net useful bit rate is:

$$R_{\text{net, 16-QAM}} = R_{\text{raw, 16-QAM}} \times R_c = 12.93 \times \frac{2}{3} \approx \mathbf{8.62 \text{ Mbit/s}} \quad (13)$$

$$R_{\text{net, 64-QAM}} = R_{\text{raw, 64-QAM}} \times R_c = 19.39 \times \frac{2}{3} \approx \mathbf{12.93 \text{ Mbit/s}} \quad (14)$$

7.4 Summary Table

Table 10: Theoretical Bit Rate Summary for Our System Parameters (4 MHz, 8K, GI=1/32, Rate 2/3)

Parameter	16-QAM	64-QAM	Unit
FFT size N_{FFT}	8192	8192	samples
Data subcarriers N_{data}	6817	6817	–
Sample rate f_s	4	4	MHz
Symbol duration T_s	2.112	2.112	ms
Bits per symbol	4	6	bit
Code rate R_c	2/3	2/3	–
Raw bit rate	12.93	19.39	Mbit/s
Net useful bit rate	8.62	12.93	Mbit/s

These values confirm that even at the reduced 4 MHz sample rate used in our implementation (instead of the standard 8 MHz DVB-T channel bandwidth), the system achieves a net throughput sufficient to carry standard-definition MPEG video streams, which typically require 3–6 Mbit/s. The 64-QAM scheme would support high-definition content, provided the hardware limitations described in Section 4 are addressed.

8. Results and Discussion

The results obtained from both simulation and real hardware experiments provide valuable insights into the performance of the DVB-T system.

Simulation vs Real Hardware

In simulation, the system operates under ideal conditions with no noise, perfect synchronization, and unlimited computational resources. As a result, the constellation diagrams appear clean and well-defined, and the transmitted data is perfectly recovered.

However, real hardware introduces several practical challenges. Noise, synchronization errors, and hardware limitations affect signal quality. Additionally, the processing capability of the host computer plays a crucial role in maintaining real-time operation.

Performance of 16-QAM

The 16-QAM modulation scheme demonstrated stable performance in both simulation and real hardware. The constellation remained well-defined, and the system successfully recovered the transmitted MPEG transport stream.

This confirms that 16-QAM provides a good balance between robustness and data rate, making it suitable for practical implementations.

Performance of 64-QAM

In contrast, the 64-QAM scheme, while offering higher data rates, showed significant limitations in real hardware. The constellation points became more dispersed due to increased sensitivity to noise and synchronization errors.

Moreover, the higher computational load led to underflow and overflow issues, preventing stable real-time transmission.

Discussion

The comparison between 16-QAM and 64-QAM highlights the trade-off between capacity and robustness. While higher-order modulation increases data throughput, it requires better channel conditions and more powerful hardware.

These results demonstrate that system design must consider both theoretical performance and practical constraints. The implementation of DVB-T using SDR is feasible, but optimization is necessary to achieve stable real-time performance.

9. Experimental Validation Tests

9.1 Test 1: 16-QAM Simulation

The system was first evaluated in simulation mode using 16-QAM modulation. The objective was to verify the correct operation of the transmission and reception chain under ideal conditions.

The results showed clean constellation diagrams and perfect recovery of the transmitted data, confirming the correctness of the implementation.

Overall Validation Summary

Table 11: Summary of Experimental Validation

Test	Success	Remarks
16-QAM Simulation	Yes	Ideal conditions, perfect decoding
16-QAM Hardware	Yes	Stable real-time transmission
Gain Adjustment	Yes	Optimal range identified
64-QAM Simulation	Yes	Correct configuration
64-QAM Hardware	No	Underflow / overflow issues
Live Streaming	Partial	Architecture validated only

Key Observation: The system achieves reliable performance under moderate complexity (16-QAM), but higher-order modulation (64-QAM) exposes limitations in both hardware processing capability and real-time constraints. This confirms that practical SDR systems must balance performance with computational feasibility.

9.2 Test 2: 16-QAM Real Hardware

The system was then implemented using the USRP 2920 hardware. The objective was to validate real-time transmission.

Stable transmission was achieved, and the output file was successfully reconstructed, demonstrating the feasibility of the system in practical conditions.

9.3 Test 3: Gain Adjustment

Table 12: Effect of Gain Adjustment on System Performance

TX/RX Gain Level	Observation	Result
Low Gain	Weak signal, noisy constellation	No decoding
Medium Gain	Stable constellation	Successful video recovery
High Gain	Signal distortion / saturation	Errors in decoding

Analysis: The system exhibits an optimal operating region where the signal-to-noise ratio (SNR) is sufficient for correct decoding without causing saturation. Too low gain leads to insufficient SNR, while excessive gain introduces nonlinear distortion, degrading the constellation quality and increasing decoding errors.

Different transmitter and receiver gain values were tested to analyze their impact on signal quality.

It was observed that excessive gain caused signal distortion, while insufficient gain resulted in weak signal reception. Proper gain tuning was essential for optimal performance.

9.4 Test 4: 64-QAM Simulation

The system was configured to use 64-QAM modulation in simulation. The results showed a well-defined constellation, confirming correct configuration.

9.5 Test 5: 64-QAM Hardware

When implemented on real hardware, the system experienced underflow and overflow issues due to increased computational requirements.

The constellation became distorted, and stable transmission could not be maintained.

9.6 Test 6: Live Streaming Attempt

A real-time transmission system using UDP streaming was designed. However, due to hardware limitations, full end-to-end validation was not achieved.

Nevertheless, the experiment demonstrated the potential for real-time DVB-T transmission using SDR systems.

10. Live DVB-T Transmission Using UDP Streaming

To extend the DVB-T system toward real-time applications, a live video transmission architecture was designed using UDP streaming. The file-based input was replaced by a UDP source, allowing continuous data transmission from an external video stream.

Transmission Flowgraph

The transmitter receives a real-time video stream via UDP (generated using FFmpeg), which is then processed through the DVB-T encoding chain including channel coding, interleaving, mapping, and OFDM modulation before transmission.

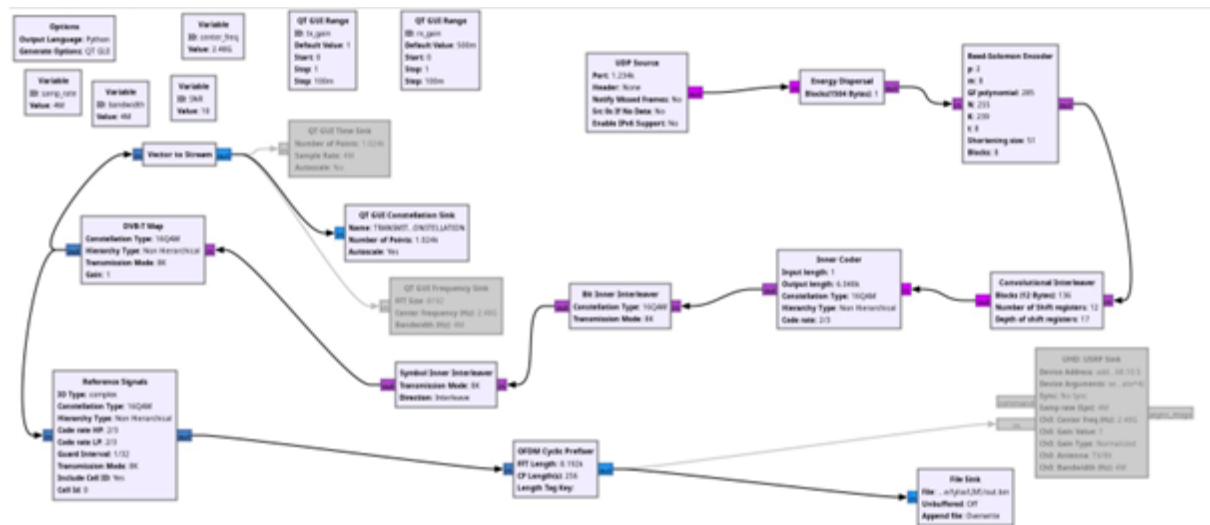


Figure 14: GNU Radio flowgraph for live DVB-T transmission using UDP streaming.

Reception Flowgraph

The receiver captures the transmitted signal using the SDR device and performs synchronization, FFT processing, demapping, deinterleaving, and channel decoding to recover the transmitted data stream.

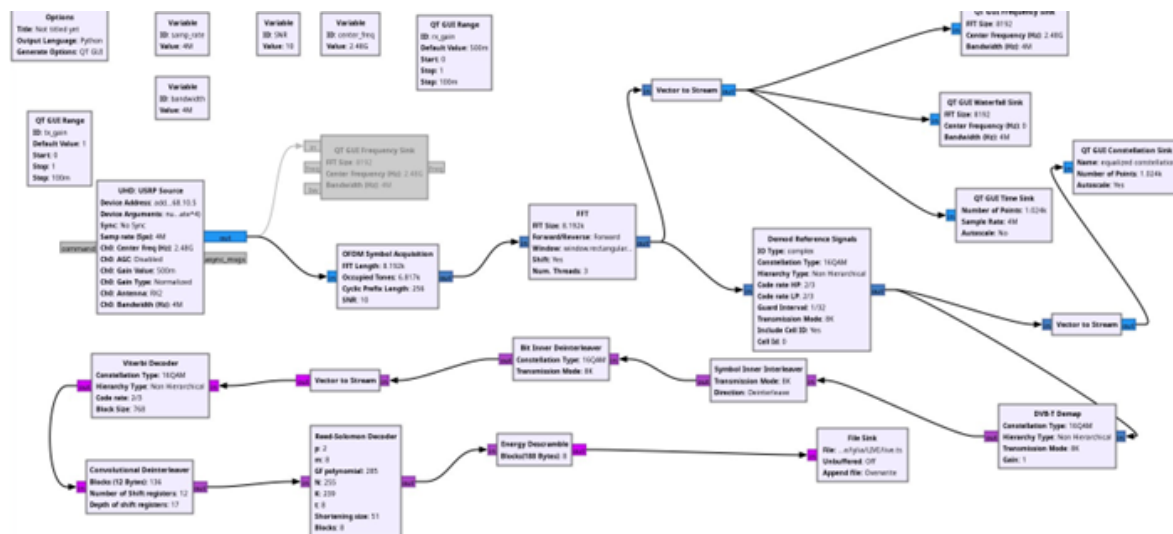


Figure 15: GNU Radio flowgraph for live DVB-T reception using UDP streaming.

Note: Due to hardware constraints – specifically the availability of a single USRP device – real-time end-to-end validation of the live transmission was not performed. Nevertheless, the proposed architecture demonstrates the feasibility of integrating real-time video streaming into an SDR-based DVB-T system.

11. Conclusion

In this work, a complete DVB-T transmission and reception chain was successfully designed and implemented using Software Defined Radio (SDR) technology and GNU Radio tools. The system was thoroughly evaluated through both simulation and real-time experiments using USRP hardware.

The results obtained with 16-QAM modulation demonstrated stable performance and reliable signal recovery, confirming the correct operation of all processing stages, from encoding to decoding. In contrast, the investigation of 64-QAM revealed the practical limitations of real hardware systems, particularly in terms of computational requirements, synchronization sensitivity, and noise tolerance.

This project highlights the trade-off between spectral efficiency and system robustness, showing that higher-order modulation schemes, while offering increased data rates, impose stricter constraints on both hardware and channel conditions.

Furthermore, the implementation of a live transmission architecture using UDP streaming illustrates the flexibility of SDR systems and their potential for real-time multimedia applications, even though full validation was limited by hardware availability.

Overall, this work provides a comprehensive understanding of DVB-T systems in both theoretical and practical contexts, emphasizing the importance of system optimization and hardware capabilities in real-world deployments. Future work may focus on improving processing efficiency, enhancing synchronization

techniques, and exploring advanced SDR platforms capable of supporting higher data rates.